

Generation Interconnection Request System Impact Study Report

For Queue Project Q16-0609

Moss Bluff 138 kV
February 16, 2018

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q16-0609. The Interconnection Customer proposes a 904.9 MW natural gas combined cycle facility interconnecting at the Moss Bluff 138 kV substation in the Kentucky area. The request was received on April 16, 2016. The study evaluated system intact and single contingency conditions using the regional planning case, including 226 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 25 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$118,256,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
Moss Bluff to Barnabie Cable Sta 138.0 kV Line	Barnabie Cable Sta to Longdon Substation 138.0 kV	212.5	452	59.2
Clubhose to Hamilton Branch 138.0 kV	Glendale S Tap 6 Substation to Campbell Junction 138.0 kV	204.8	232	29.5
N Hendrickson to School Street 138.0 kV Line	Glendale S Tap 6 to Campbell Jct 138.0 kV	204.1	209	24.5
Hamilton Branch to Universal Studios Tap 138.0 kV	Cmpbell Junction to Glendale S Tap 6 138.0 kV line	201.3	234	29.5
Lovett to Idaho Falls Power 138.0 kV Line	Casino to Barnabie Cable Sta 138.0 kV	197.2	349	37.1
Anson N to Biscoe 138.0 kV	Cedar Tap to Terry Co Substation 138.0 kV L	193.8	357	36.7
Glendale S Tap 6 to Biscoe 138.0 kV	Cedar Tap to Terry Co 138.0 kV	187.5	378	36.7
N Hendrickson to Clubhouse 138.0 kV	Glendale South Tap 6 Substation to Campbell Junction 138.0 kV L	185.3	271	29.5
Rapids Energ Center to Casino 138.0 kV	Cedar Tap to Terry Co 138.0 kV line	182.9	587	54.4
Glendale South Tap 6 to Campbell Jct 138.0 kV	Cedar Tap to Terry Co 138.0 kV	180.9	402	36.7
Casino to Barnabie Cable Sta 138.0 kV L	Cedar Tap to Terry Co 138.0 kV	180.8	599	54.4
Terry Co to Skeena 138.0 kV line	Casino to Barnabie Cable Sta Substation 138.0 kV	177.1	557	48.9
Terry Co Substation to Cedar Tap 138.0 kV	Barnabie Cable Sta to Casino 138.0 kV Line	176.3	562	48.9

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
Fernwood to Chestnut 138.0 kV	Chestnut to Consolidated Coal 138.0 kV L	175.7	249	21.5
Campbell Jct to School Street 138.0 kV	Glendale South Tap 6 to Campbell Jct 138.0 kV line	174.0	302	29.6
Lovett to Barnabie Cable Sta 138.0 kV	Barnabie Cable Sta to Casino 138.0 kV	171.5	359	33.8
Rancho Carmel to Lanham Substation 138.0 kV	Albion Rd to Platte City 138.0 kV	169.6	238	20.0
Idaho Falls Power to Cedar Tap 138.0 kV	Barnabie Cable Sta to Casino 138.0 kV	165.9	420	37.1
Sand Springs to C-33 Doe 138.0 kV	Davisville to N Appleton 138.0 kV	164.4	390	25.0
Platte City to C-33 Doe 138.0 kV Line	N Appleton to Davisville 138.0 kV line	162.2	400	25.0
Wixom to Ayers Island 138.0 kV	Davisville to N Appleton 138.0 kV	162.1	410	33.2
Consolidated Coal to Chestnut 138.0 kV Line	Fernwood to Chestnut 138.0 kV	161.4	242	18.3
Campbell Jct to Rapids Energy Center 138.0 kV	Cedar Tap to Terry Co 138.0 kV L	160.0	659	54.4
Albion Road to Platte City Substation 138.0 kV line	Davisville to North Appleton 138.0 kV	159.5	413	25.0
Saint Joseph to Chestnut Substation 138.0 kV Line	Wixom to Albion Rd 138.0 kV	157.7	429	29.5

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Moss Bluff to Barnabie Cable Sta 138.0 kV Line	Rebuild line with higher-capacity conductor	\$8,282,000	61% (\$5,042,000)
Clubhose to Hamilton Branch 138.0 kV	Rebuild line with higher-capacity conductor	\$19,253,000	48% (\$9,260,000)
N Hendrickson to School Street 138.0 kV Line	Rebuild line with higher-capacity conductor	\$2,279,000	51% (\$1,158,000)
Hamilton Branch to Universal Studios Tap 138.0 kV	Rebuild line with higher-capacity conductor	\$8,683,000	48% (\$4,177,000)
Lovett Substation to Idaho Falls Power Substation 138.0 kV L	Rebuild line with higher-capacity conductor	\$6,194,000	52% (\$3,194,000)
Anson North to Biscoe 138.0 kV	Rebuild line with higher-capacity conductor	\$3,857,000	51% (\$1,961,000)
Glendale S Tap 6 to Biscoe 138.0 kV	Rebuild line with higher-capacity conductor	\$2,324,000	51% (\$1,182,000)
N Hendrickson to Clubhouse 138.0 kV	Rebuild line with higher-capacity conductor	\$2,509,000	48% (\$1,207,000)
Rapids Energ Center to Casino 138.0 kV	Rebuild line with higher-capacity conductor	\$2,270,000	50% (\$1,129,000)

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Campbell Junction to Glendale S Tap 6 138.0 kV	Rebuild line with higher-capacity conductor	\$4,919,000	51% (\$2,501,000)
Barnabie Cable Sta to Casino 138.0 kV L	Rebuild line with higher-capacity conductor	\$13,886,000	50% (\$6,906,000)
Skeena to Terry Co 138.0 kV	Rebuild line with higher-capacity conductor	\$13,003,000	50% (\$6,482,000)
Terry Co Substation to Cedar Tap 138.0 kV	Rebuild line with higher-capacity conductor	\$8,147,000	50% (\$4,061,000)
Fernwood to Chestnut 138.0 kV	Rebuild line with higher-capacity conductor	\$9,530,000	37% (\$3,520,000)
School St to Campbell Jct 138.0 kV	Rebuild line with higher-capacity conductor	\$9,099,000	48% (\$4,377,000)
Lovett to Barnabie Cable Sta 138.0 kV	Rebuild line with higher-capacity conductor	\$21,244,000	52% (\$10,964,000)
Rancho Carmel to Lanham Substation 138.0 kV	Rebuild line with higher-capacity conductor	\$7,511,000	39% (\$2,932,000)
Cedar Tap to Idaho Falls Power 138.0 kV	Rebuild line with higher-capacity conductor	\$6,480,000	52% (\$3,341,000)
Sand Springs to C-33 Doe 138.0 kV	Rebuild line with higher-capacity conductor	\$11,208,000	36% (\$4,096,000)
Platte City to C-33 Doe 138.0 kV Line	Rebuild line with higher-capacity conductor	\$7,979,000	36% (\$2,916,000)
Wixom to Ayers Island 138.0 kV	Rebuild line with higher-capacity conductor	\$24,974,000	42% (\$10,517,000)
Consolidated Coal to Chestnut 138.0 kV	Rebuild line with higher-capacity conductor	\$34,945,000	41% (\$14,219,000)
Campbell Jct to Rapids Energy Center 138.0 kV	Rebuild line with higher-capacity conductor	\$3,998,000	50% (\$1,988,000)
Albion Road to Platte City 138.0 kV	Rebuild line with higher-capacity conductor	\$26,204,000	36% (\$9,576,000)
Saint Joseph to Chestnut Substation 138.0 kV Line	Rebuild line with higher-capacity conductor	\$4,216,000	37% (\$1,550,000)
	Total allocated to project		\$118,256,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.